

On the origin of the asymmetry of ground densities in inclined showers

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1 Introduction

In a previous GAP note [1], Cl. Pryke pointed out a large asymmetry of the ground densities obtained from simulated inclined showers: at a given distance from the shower axis, the density of particles (especially photons and electrons) is found to be higher in the “early” region (on the side below the core) than on the “late” region (on the opposite side). He proposed an interpretation based on the decrease of the shower after the maximum of its longitudinal development.

In this study we want to show that, at moderate zenith angles (typically less than 40°), at the altitude of the AUGER sites, this longitudinal decrease cannot contribute to a significant part of the effect (Section 2). We give a geometrical interpretation of the asymmetry, based on the radial divergence of the particles from the axis (Section 3 and 4). Then we check our model with both vertical and inclined simulated showers (Section 5), and derive the actual asymmetry expected on the tanks signals (Section 6).

2 Contribution of the longitudinal evolution

The average ground level in the AUGER sites corresponds to an atmospheric depth between 870 and 890 g.cm^{-2} , that is close to the value of X_{max} expected for proton induced showers in the range 10^{19} to 10^{20} eV. If we consider the longitudinal development at large distances from the core, we can expect that the electromagnetic densities begin to decrease at a depth slightly higher than X_{max} (which reflects essentially the maximum of the activity of the core): at zenith angles up to 40° , the slant depth is less than 1150 g.cm^{-2} , and the rate of attenuation (if any) should remain moderate.

We have checked this point with sets of 20 showers (in order to reduce the incidence of shower-to-shower fluctuations) induced by vertical protons at 10^{19} and 10^{20} eV, using AIRES/SIBYLL version 2.2.1 (with a slight modification to allow the ground to be below the sea level). We have extracted the ground particles at 900 , 1000 , 1100 and 1200 g.cm^{-2} .

Figure 1 shows the longitudinal evolution of the electromagnetic and muonic densities, compared to a “reference” depth (900 g.cm^{-2}). The electromagnetic density is attenuated by a factor around 15% per 100 g.cm^{-2} (i.e. per km of longitudinal development at the altitude of the site); the muonic density is *increasing*,

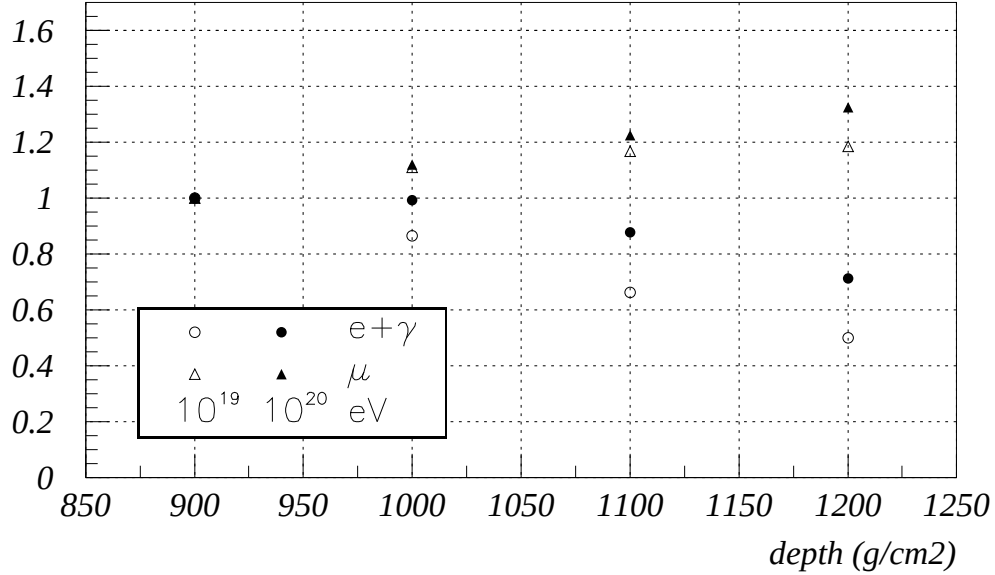


Figure 1: Evolution of the electromagnetic and muonic densities with the depth, in vertical showers, using 900 g.cm^{-2} as a reference

at least up to 1200 g.cm^{-2} (of course, the total number of muons decreases, but their spread increases).

Above 10^{20} eV , we can reasonably assume that the electromagnetic attenuation is negligible up to 1000 g.cm^{-2} , and less than 15% per km up to a depth corresponding to a zenith angle $\theta = 30^\circ$. At this zenith angle, an “early” and a “late” point at $r = 500 \text{ m}$ differ by $2r \sin \theta = 500 \text{ m}$ in longitudinal coordinate: the asymmetry due to attenuation should not exceed $\pm 10\%$; at larger distances, the contribution of muons reduces, and possibly reverses, the asymmetry related to the longitudinal evolution. This longitudinal evolution cannot therefore explain the asymmetries observed on the ground densities.

3 Basic understanding of the geometric effect

One can easily understand that there should be a “early”/“late” asymmetry in the ground signal by considering figure 2. A purely symmetric event in the shower frame (2 particles going away from the axis with the same angle α) results in a non symmetric effect in the ground plane (only the particle in the “early” region can be seen at ground). When projecting the ground densities onto the shower plane, as the direction of individual particles is not known, one has to use the shower direction. This results in what we call a geometric asymmetry which can be seen on figure 3. The particles hitting the ground in the “early” region will be seen more vertical than the ones in the “late” region. Even with a perfect symmetry around its axis and an invariance by translation, an inclined shower will not give symmetric patterns on the ground.

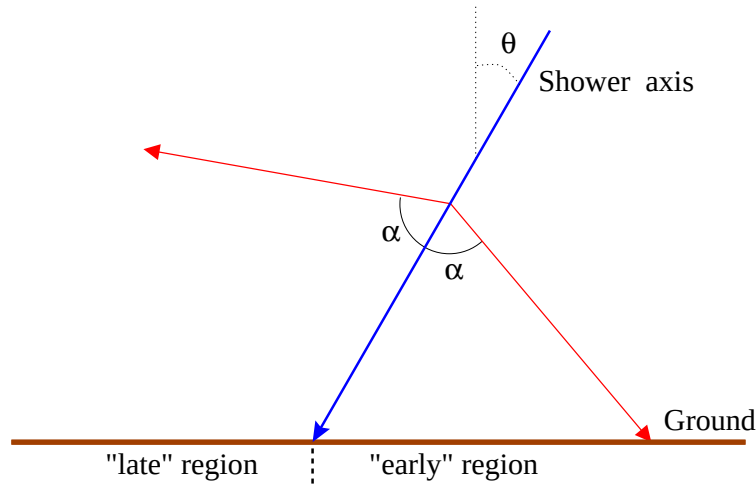


Figure 2: 2 particles going away from the shower with the same angle α will have different behavior due to the θ inclination of the shower.

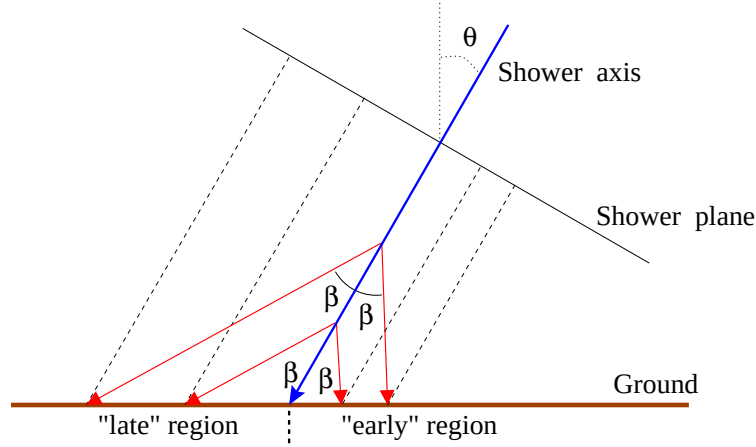


Figure 3: Particles going away from the shower axis with the symmetric angles β . When projecting the ground densities on the shower plane, a purely geometric asymmetry appears.

4 Mathematic Model

In this section, we develop some calculations of the asymmetry. We use simplified formulas to give a better understanding of the effect we expect.

We consider the distribution of momenta unit vectors of particles reaching the ground in a vertical shower. In the shower frame, with cylindrical coordinates, we note $N(r, \Phi, z)[r; \Phi; z; p_r; p_\Phi; p_z]$ this distribution, $N(r, \Phi, z)$ being a global normalization factor, (p_r, p_Φ, p_z) the normalized projections on the 3 axis (with $p_z = -\sqrt{1 - p_r^2 - p_\Phi^2}$ as particles are going downward).

We assume in a first approximation that the shower is stationary, *ie* that the development of the shower is negligible, as was seen in section 2 (shower invariant through translations along the z axis), and has a cylindric symmetry. The distribution of particles impulsions only depends then on the distance r to the axis. Let us also assume that the orthoradial divergence is negligible compared to the radial divergence: $p_\Phi \ll p_r$.

When a shower is analyzed, what is usually computed is the ground density of

particles ρ . This is actually the flux of particles going through the ground, *ie* the distribution of $-P_z$. In our simple model, we have (ρ_v stands for vertical):

$$\rho_v(r) = \langle -p_z(r) \rangle N(r) \quad (1)$$

with $\langle X \rangle$ the average of X .

If we make a θ rotation around some $\vec{y}$ axis, the distribution of impulsions is now depending on r and Φ . Some algebra on a vector $\vec{P} = p_r \vec{e}_r + p_z \vec{e}_z$ in cylindric coordinates leads in cartesian coordinates system (e_x, e_y, e_z) to:

$$\begin{aligned} \vec{P}(r, \Phi) = & \left(p_r(r) \cos \Phi \cos \theta + p_z(r) \sin \theta \right) \vec{e}_x + p_r(r) \sin \Phi \vec{e}_y \\ & + \left(-p_r(r) \cos \Phi \sin \theta + p_z(r) \cos \theta \right) \vec{e}_z \end{aligned} \quad (2)$$

The ground density after projection in the shower plane (ρ_θ stands for inclined, computed from $-P_z / \cos \theta$) is now:

$$\rho_\theta(r, \phi) = \left(p_r(r) \cos \Phi \tan \theta - p_z(r) \right) N(r) \quad (3)$$

Comparing (3) to (1) leads to an asymmetry factor ($\rho_\theta / \rho_v - 1$):

$$\mathcal{A}(r, \phi) = \mathcal{A}(r) \cos \Phi = \left\langle \frac{p_r(r)}{-p_z(r)} \right\rangle \tan \theta \cos \Phi \quad (4)$$

The average of the quantity $p_r(r)/p_z(r)$ must be carefully computed as only particles having $P_z < 0$ (equivalent to $p_z(r) < p_r(r) \cos \Phi \tan \theta$) must be taken into account ($P_z > 0$ being upward going particles).

Anyway, the purely geometric asymmetry expected is roughly a $\mathcal{A}(r) \cos \Phi$ asymmetry, $\mathcal{A}$ depending on r as a function of the distribution of radial (and orthoradial in a more complex model) impulsions.

5 Monte Carlo check of the model

Here we assume that the development of the electromagnetic component of an inclined shower is the same as in a vertical shower at the same cumulated atmospheric depth (actually there is a slight anisotropy because the density of air is not uniform in a plane perpendicular to the shower axis). Then, if the ground corresponds to the atmospheric depth X_g , a shower with zenith angle θ is comparable to a vertical shower at $X_g / \cos \theta$, rotated by θ .

If the asymmetry is purely geometric as we claim, then we must be able to reproduce it with a vertical shower at the modified depth. For that, we rotate the vertical shower by a θ angle and project each particle on the rotated ground *along its own direction*, not along the shower axis. We obtain projected densities on the ground. In all this section, we apply this method to the electrons and photons only as they show a more important asymmetry than muons. Muons are just ignored for all the computations.

Figure 4 shows the comparison of the projected densities to the direct observation of an inclined shower on a ground at 900 g.cm^{-2} , in different slices in r . At 30° the asymmetry is very well reproduced; at 40° , the longitudinal attenuation begins to play a role at large distances, but the geometrical effect is still predominant.

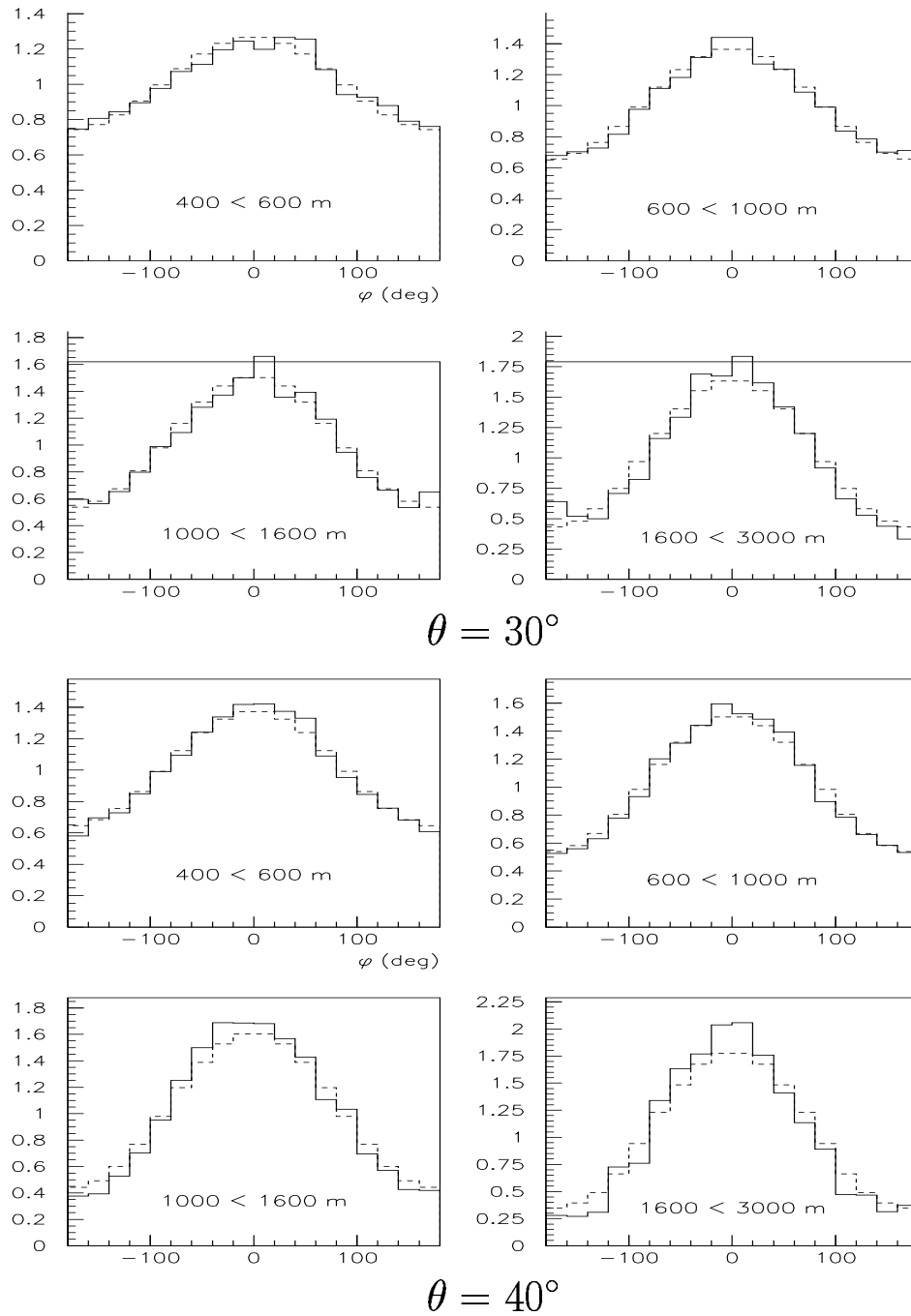


Figure 4: Comparison of the ground asymmetry (electromagnetic energy density as a function of Φ angle) observed in a simulated inclined shower (solid lines) to the prediction of a vertical shower at the same depth, after projection (dashed line)

Figure 5 shows that the radial divergence of the particles increases with r : it is relatively moderate for muons (because they come roughly in straight line from the end of the hadronic cascade, at an altitude around 5 km above the ground), while the photons and electrons are much more divergent.

One can then extract from a set of 20 vertical showers with the ground at 900 g.cm^{-2} the mean value $\mathcal{A}(r)$ appearing in equation 4 from figure 5 and compare it to a $\cos \Phi$ fit on the inclined shower. The results are in table 1. The approximation of section 4 is in quite good agreement with the data at 30° . At

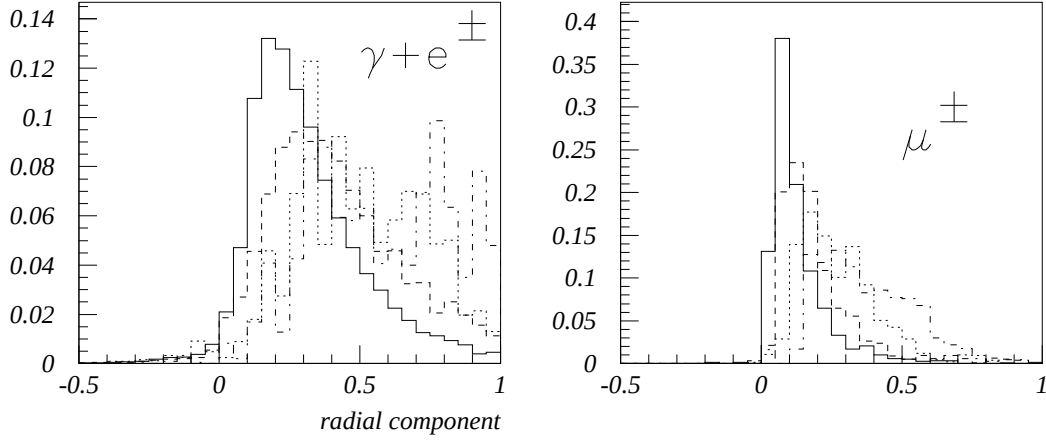


Figure 5: Radial component p_r of particles in a vertical shower. Solid: $300 < r < 500$ m; dashed: $500 < r < 1000$; dotted: $1000 < r < 1500$; dash-dotted: $r > 1500$

r	$\mathcal{A}(r)$	fit	r	$\mathcal{A}(r)$	fit
$400 < 600$	0.25	0.24	$400 < 600$	0.35	0.39
$600 < 1000$	0.34	0.34	$600 < 1000$	0.45	0.50
$1000 < 1600$	0.45	0.47	$1000 < 1600$	0.56	0.65
$1600 < 3000$	0.57	0.61	$1600 < 3000$	0.67	0.77
$\theta = 30^\circ$			$\theta = 40^\circ$		

Table 1: Asymmetry $\mathcal{A}(r)$ computed from a set of 20 vertical 10^{20} eV shower with formula from equation 4 at 30° and 40° compared to a $\cos \Phi$ asymmetry fit on 10^{20} eV simulated inclined shower with these angles.

40° , the longitudinal extinction of the shower becomes visible.

We can conclude that most of the electromagnetic ground densities asymmetry observed at moderate angles comes from the geometric effect. This asymmetry can be quite important (in table 1, we see a $\pm 50\%$ at 2 km for a 30° shower, that is a factor of 4 between the “early” and “late” regions), but can be well estimated by a simple model. Of course, the result is also valid for muons, although less spectacular.

6 Asymmetry effect on an Auger Detector

Until now, we have considered only ground densities. However, the Auger detectors are not scintillators but Čerenkov tanks, and some part of the signal comes from particles entering the side walls of the tank. One can easily understand that the geometric effect described above will give an asymmetry in the side signals opposite to the one from the top, as the walls are orthogonal to the ground. When we expect less signal to come from the top, we must also expect more signal to come from the side. Equation (2) gives some insight on the expected asymmetry, which is more complex:

$$\mathcal{A}_{side} = \sqrt{\left(\frac{p_z}{p_r}\right)^2 \sin^2 \theta - \cos^2 \Phi \sin^2 \theta + \frac{p_z}{p_r} \cos \Phi \sin 2\theta}$$

As usually $p_z < 0$ and $p_r > 0$, we can expect an *increase* in the signal from the $(\frac{p_z}{p_r} \cos \Phi \sin 2\theta)$ term for $\Phi \approx \pi$. Therefore, the “late” region is enhanced with respect to the “early” one.

In simple terms, top entries are favored in the “early” region, whereas side entries are more common in the “late” region. We can expect that:

- the electrons and photons will give a strong asymmetry. As the top surface of an Auger tank is 2.4 times larger than the side surface, the top signal will prevail over the side one, and we expect the electromagnetic signal to be higher in the “early” region than in the “late” one;
- the muons on the other hand will give a smaller asymmetry. But as the signal of a muon is roughly proportional to the length of water crossed, the side signal per muon will be much stronger than the top one (the factor 2.4 being reversed). This compensates the surface factor, and we expect the “early” muonic signal to be the same as the “late” one.

Figure 6 shows the signal asymmetries for a vertical 10^{20} eV shower inclined by 30° . As the muonic part of the signal becomes more and more important (at large distance from the core), the asymmetry is compensated.

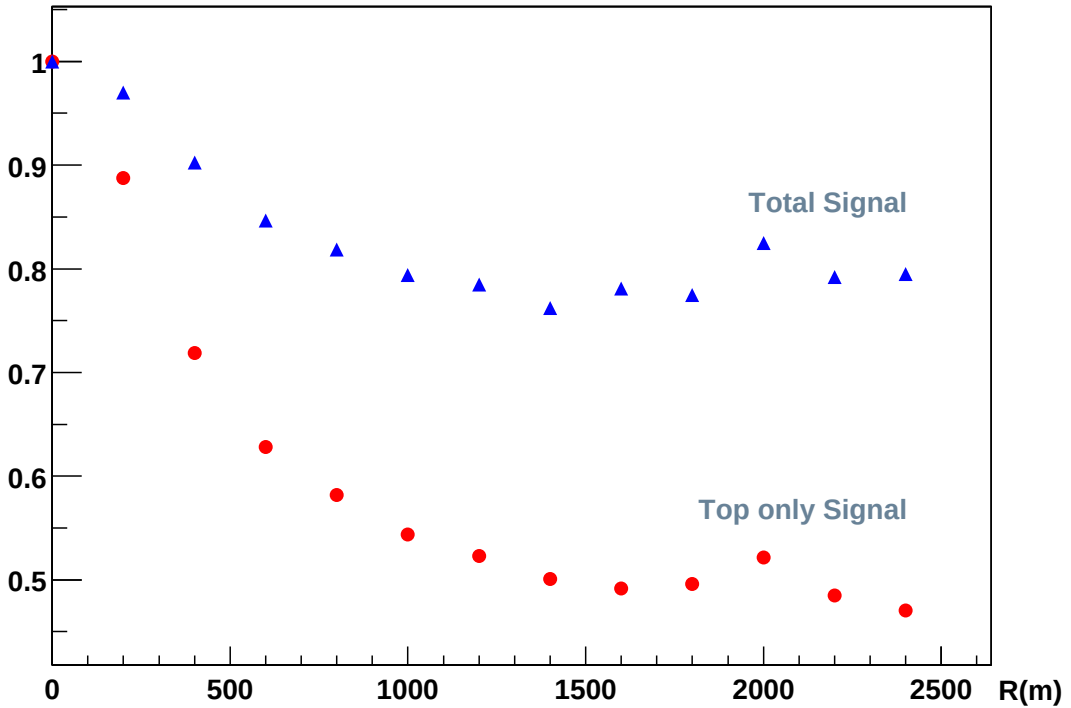


Figure 6: Signal “late” to “early” geometrical asymmetry for a vertical 10^{20} eV shower inclined by 30° , as a function of distance to axis. The large asymmetry for ground densities (Top signal) is greatly reduced by the side signal to give a total asymmetry of about $\pm 10\%$ at more than 2 km.

7 Influence on the reconstruction

We’ve seen in the previous section that the geometric asymmetry appears to be important on ground densities but is not as important on the total signal due to particles entering the sides of the tanks. We found $\pm 10\%$ at 30° , and $\pm 15\%$ at 40° .

These values are comparable with the attenuation seen in Section 2 (≈ 0 at 30° and $\pm 15\%$ per longitudinal km at 40° , that is $\pm 10\%$ per km on the ground). The asymmetry due to the longitudinal attenuation may even be stronger with older showers, which is the case of QGSJET showers. We’ve seen that this attenuation only affects the electromagnetic part (at moderate zenith angles) and that muons show an *increase* in density. We can expect the signal asymmetries to be like the ones in figure 6, more marked at medium distances (where the electromagnetic part dominates), and less visible at large distance (where the signal comes from muon spreading).

We checked that and obtained a maximum asymmetry between 1 and 1.5 km. With extremely old showers (due to a bug in AIRES 2-2-0 version of QGSJET), it can be up to $\pm 30\%$ at 40° . However, even with these large values, the current reconstruction results are not affected. The only bias is on the position of the core which is shifted a few tens of meter in the “early” region. This systematic bias can be easily corrected.

8 Conclusion

The asymmetry of ground densities in inclined showers is shown to be dominated by a purely geometrical projection effect. This effect can be quite important, leading to a ratio “early” to “late” density of up to 5 for the electromagnetic part of a 40° shower (and 2.5 for the muonic part) at 3 km from the core.

However, the side signals from the Auger Čerenkov tanks compensate for a large part this asymmetry. Even with the additional asymmetry due to the longitudinal attenuation, the total signal asymmetry is found to be less than a factor of 1.4 at 40° (increasing to 1.8 with the bugged QGSJET of AIRES 2-2-0 version). We are quite confident in these figures, and we believe they differ from [1] mainly because the tank simulation program used in [1] didn’t take into account individual particles direction as our sampling algorithm does.

The Reconstruction algorithm showed to be very robust on these inclined showers, and the only bias found was a limited shift in core position. This has been corrected, as it is important for the Hybrid reconstruction to have a precise core location.

Most of all, the different contributions to the signal asymmetry seem to be well understood.

References

- [1] Cl. Pryke: GAP 98-034, Asymmetry of Air Showers at Ground Level